

Software Test Plan (STP) - File Conversion Service

Project: File Conversion Service API

Version: 1.0

Authors:

TEAM NAME : CONVERTY

- Anivarth Upadhyaya PES1UG23CS079
- Ananya Joshi PES1UG23CS068
- Ayush Gowda PES1UG23CS124

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Status: Sample / Draft

1. Introduction

Purpose: This document defines the test plan for the File Conversion Service v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers API endpoints and service features such as file upload, format conversion (e.g., PDF, DOCX, CSV, JSON, Markdown), error handling, and response performance. Core storage systems and third-party converters not owned by the team are excluded.

References: API Specification v1.0, Design Documentation v1.0, File Format Standards (PDF, CSV, JSON, DOCX, MD).

Definitions: API (Application Programming Interface), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix).

2. Test Items

- File upload module
- Format conversion engine (PDF, DOCX, CSV, JSON, Markdown)
- Error handling and reporting

- Authentication/authorization module (if applicable)
- Logging and monitoring module
- Admin/maintenance interface

3. Features to be Tested

Requirement ID	Feature Description
FCS-F-001	Upload file for conversion
FCS-F-010	Convert file from one format to another (Markdown → PDF, CSV → JSON, etc.)
FCS-F-020	Download converted file
FCS-F-030	Validate unsupported/invalid file formats
FCS-NF-001	Response time ≤ 3s for small files (≤5MB)
FCS-NF-002	Accuracy of conversion ≥ 99%
FCS-NF-003	Service uptime 99.9%

4. Features Not to be Tested

- Internal file format parsing libraries (tested by vendor)
- External storage systems (e.g., AWS S3, GCP Buckets)
- Network-level latency beyond service boundary

5. Test Approach / Strategy

Levels:

- Unit tests (module-level, conversion utilities)
- Integration tests (API endpoints ↔ storage backend)
- System tests (end-to-end file conversion flows)
- Acceptance tests (UAT)

Types:

- Functional testing (conversion, error handling)
- Regression testing
- Performance testing (conversion speed, load handling)
- Security testing (file upload restrictions, input validation)
- Usability testing (API documentation clarity)

Entry Criteria: Stable build delivered, API endpoints deployed, test data prepared.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate input sanitization (reject executable/malicious files)
- Ensure TLS 1.2+ for secure communication
- Token-based authentication for API requests
- Fuzz testing for file uploads
- Penetration testing of API endpoints

6. Test Environment

Hardware: Test server with sufficient storage ($\geq 10\text{GB}$) and memory ($\geq 8\text{GB}$).

Software: File Conversion Service v1.0, mock storage backend, API gateway.

Tools: Postman (API testing), JMeter (performance), Selenium (if UI exists), Jira (defect tracking).

Test Data: Sample files in multiple formats (DOCX, PDF, CSV, JSON, Markdown).

7. Test Schedule

Milestone	Date
Test case design	12-Sep-2025
Environment setup	14-Sep-2025
Test execution start	15-Sep-2025
Test execution end	25-Sep-2025
UAT	27-Sep-2025 to 30-Sep-2025

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Ayush Gowda	Prepare plan, coordinate execution
Test Engineer	Anivarth U	Design & execute test cases, log defects
Developer	Anivarth U	Support defect fixes and triage
Product Owner	Ananya Joshi	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Maintain backup environment on cloud VM
Dependency on third-party file format libraries	Engage vendor early and maintain test stubs

11. Assumptions & Dependencies

- API specifications will be stable and available
- Test data (sample files) will be provided before execution
- External libraries for parsing/formatting will be available

12. Suspension & Resumption Criteria

Suspend testing if:

- API unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)

Resume testing if:

- Blocking defects are resolved
- API stabilized

13. Test Case Management & Traceability

Requirement ID	Mapped Test Cases
FCS-F-001 (File upload)	TC-UP-01, TC-UP-02
FCS-F-010 (File conversion)	TC-CONV-01, TC-CONV-02

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Ayush Gowda	
Dev Lead	Anivarth U	
Product Owner	Ananya Joshi	